CARLETON UNIVERSITY · AUTONOMOUS VEHICLE CAPSTONE

AAV 0.2: sim-complete, car-ready

What's in release 0.2, by area, with pictures. It carries on from What's new (22 October to 1 November), which stays as history.

The short version

- **0.2 = sim-complete, car-ready.** Everything runs in Gazebo and every test passes. **Nothing has moved the real car yet**: that's 0.3.
- **Screens:** a hardware panel that can find the steering's ends; a cockpit with every driving detail on one screen; new colours everywhere; a little robot hiding in them.
- **Car day:** in two days. Day A destroys nothing; Day B upgrades, once the backups are proven.
- **Commands:** 89 → 45. One page lists them all.
- **New on 2 Nov:** the diffusion planner as a trial (`planner:=diffusion`, off by default), Initial D on every screen, the joystick-alive check.
- **Ideas kept for 0.3:** the talking model.

Status words: **sim** = works in Gazebo; **built** = built and tested, not in a full run; **planned** = a design, no code.

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0.2 in one page

Area	What 0.2 has	Status
Steering	find the steering actuator's ends and centre: limp or jog, mark, equal limits, a lock-to-lock check that stops on resistance	built
Screens	hardware panel (2 pages), cockpit, debug screen, HUD, map; Initial D red and black on every screen (Catppuccin's still a yaml line away); the little robot	sim
Car day	Day A and Day B, the people, sign-offs, a setup script with STOPs	written; dry-run only
Know the car	the tests that measure the car, in order, with when to stop	written
Commands	45, grouped	done
Localization, perception, planning	as in What's new; the lattice is the default planner where the car may reverse	sim
Diffusion planner	a trial: a model draws the path, the lattice guards (`planner:=diffusion`, off)	sim

When is 0.2 done?

When every box on the 0.2 checklist is ticked. Where it stands on 2 November:

Check	State
Every test green (`pixi run test`)	✅ 1516 tests, 0 failures (2 Nov)
The kit ready (`car-day-check`)	⬜ 11 missing: downloads and the ESP32 tools (below)
The ESP32's emulator check (`esp qemu-check`)	⬜ the ESP32 tools not installed on this Mac yet
`car_setup.sh`'s dry run, every role	✅ only the two planned refusals
The first-move gaps (TODO 74)	🔧 the joystick-alive check built, passed on the bench (2 Nov), its car test in A9b; ⬜ the throttle cap
The people named	⬜ Ben and the team

Check	State
The hardware lead's answers (H1, C1, `.211`, the router)	the hardware lead

What came in on 1 and 2 November

What	Where	Status
The steering calibration, from the firmware to the screen	below	built
The cockpit's first screenshot, live from Gazebo	below	sim
Debug screen and hardware panel in the new colours	below	sim
Car day in two days, the people, sign-offs	below	written
`car_setup.sh`: set up one computer	Car setup	dry-run only
Know the car: the order of the tests, and when to stop	Know the car	written
89 commands down to 45	below	done
The little robot (a tribute to WALL-E)	below	built
The diffusion planner	below	sim
The road to 0.2, reviewed by three reviewers	design	done

The commands

Less is more: 89 commands became 45. Each group is one command with a word after it.

Group	Commands
Every day	`build`, `test`, `lint`, `docs`
Try it without the car	`turtle`, `sim`, `bench`, `demo` (`--debug`), `pick`, `keyboard`, `view`, `carla-car`, `drive-to-point`, `ghost`, `sim-fault`
The screens	`hud`, `map`, `debug-hud`, `hw-panel` (each with `--demo`), `rviz`, `graph`

Group	Commands
Check that it works	`check <profile>`, `bench-check`, `sim-check`, `measure <what>`, `ndt-map`
At the car	`car <machine>`, `estop`, `clear <estop, gate, geofence, watchdog>`, `actuator-test`, `calibrate-camera`, `doctor`
Before car day	`car-day-check`, `car-day-build`, `car-kit`, `get <what>`
Firmware	`pico <board> <step>`, `esp <step>`
AI and maps	`detect`, `laptop-bench`, `talk`; `-e mcp mcp`, `-e routes router`

Old	New
`hud-demo`, `debugdemo`	`hud --demo`, `demo --debug`
`estop-clear`, `gate-clear`	`clear estop`, `clear gate`
`esp-build`, `spine-flash`	`esp build`, `pico spine flash`
`get-models`, `vendor`	`get models`, `get vendor`
`controller-compare`, `timing-test`	`measure controllers`, `measure timing`

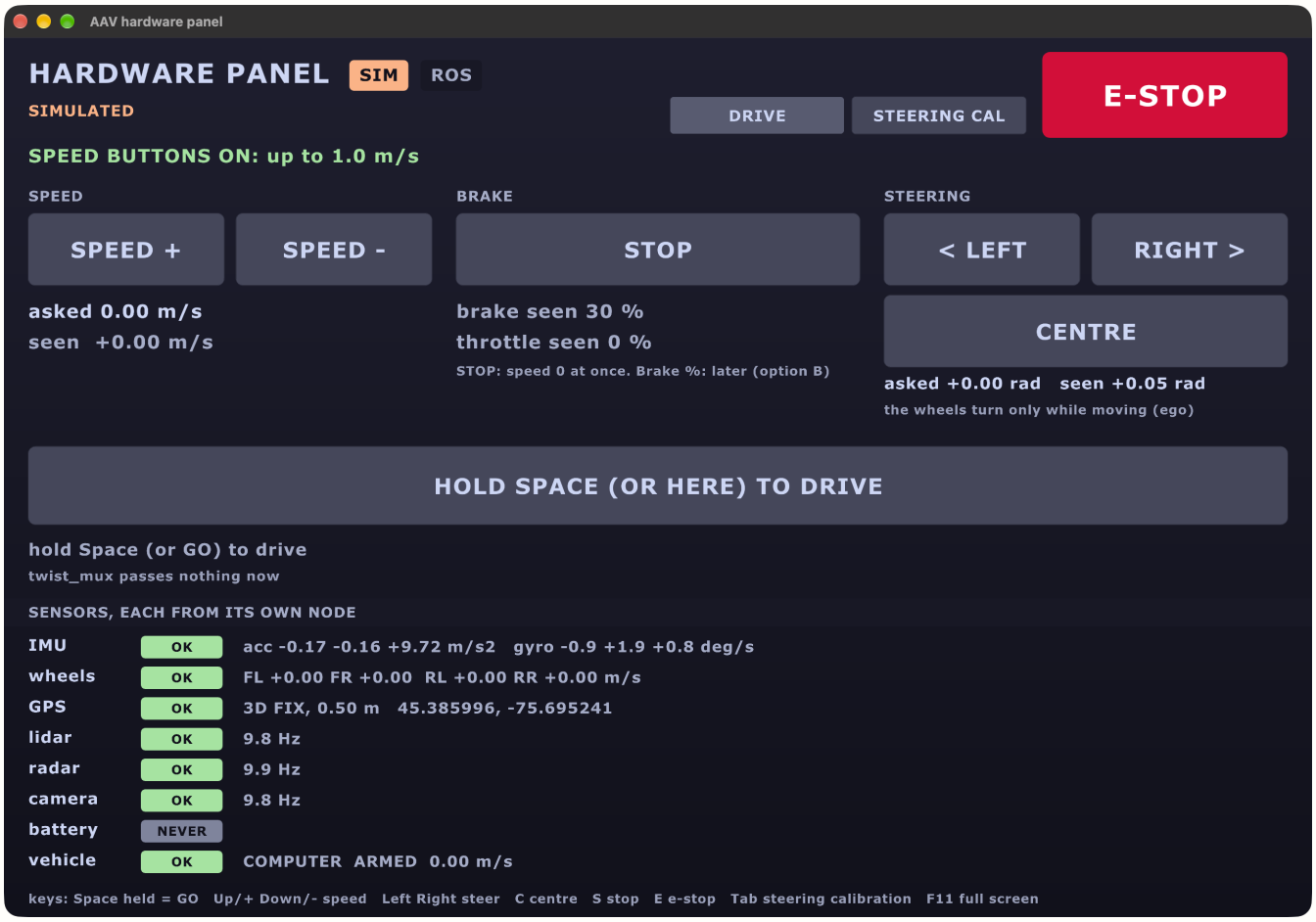
Every command, with what it does: README's "Every command". The rare ones run as scripts (its "Without a task" table).

The screens

Screen	Where	For
Hardware panel	a laptop, wired	the hardware at the lowest level: buttons, sensors, the steering's ends
Cockpit (Lichtblick): the Drive screen	the lab or a laptop	every driving detail on one screen, and which algorithms run
Debug screen	the lab or a laptop	what the car sees, in 3D
HUD and **map**	in the car	speed, state, the route

The hardware panel

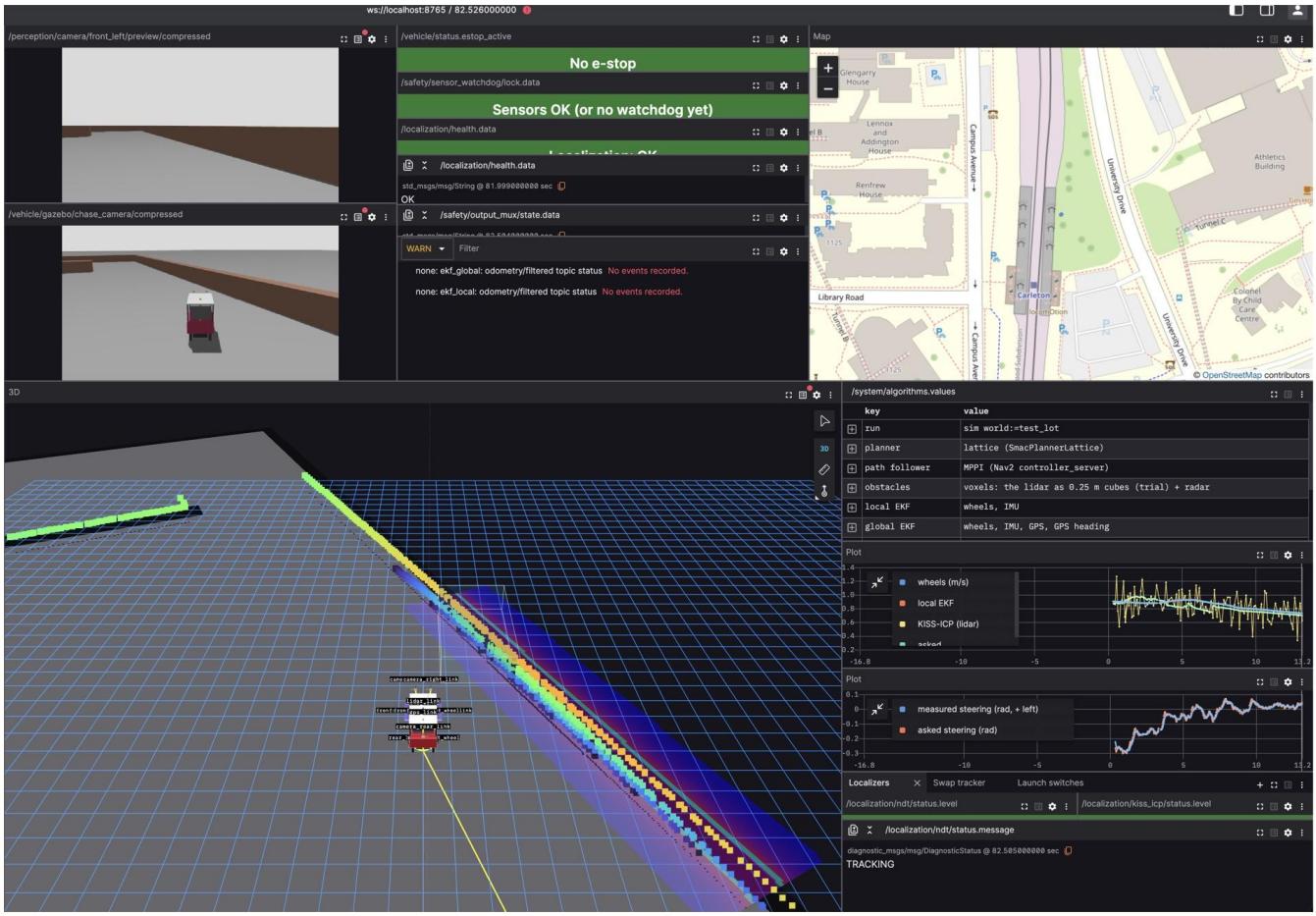
Every sensor from its own node, and very simple buttons. The car moves only while you **hold** Space. Live against Gazebo:



Its second page, **STEERING CAL**, is below.

The cockpit: the Drive screen

Every driving detail on one screen (Ben, 3 Nov: "the gazeboo with nav2 selection with cubes and stuff all as a single screen"): both cameras, the status lights, the street map, the 3D view with the cubes, the plan and **each localizer**, the speed and steering plots, and a new **Algorithms** table: which planner, path follower and localizers this run uses. Live in Gazebo's test lot, 3 Nov:



Panel	New on 3 Nov
Algorithms	planner, path follower, obstacles, the EKFs' inputs, NDT, KISS-ICP, AMCL (`/system/algorithms`)
3D	`/map` ; the EKFs' poses, NDT's and AMCL's; NDT's scan
Speed plot	KISS-ICP's speed and the local EKF's beside the wheels'
Tabs	Localizers (NDT, KISS-ICP lights) · Swap tracker · Launch switches

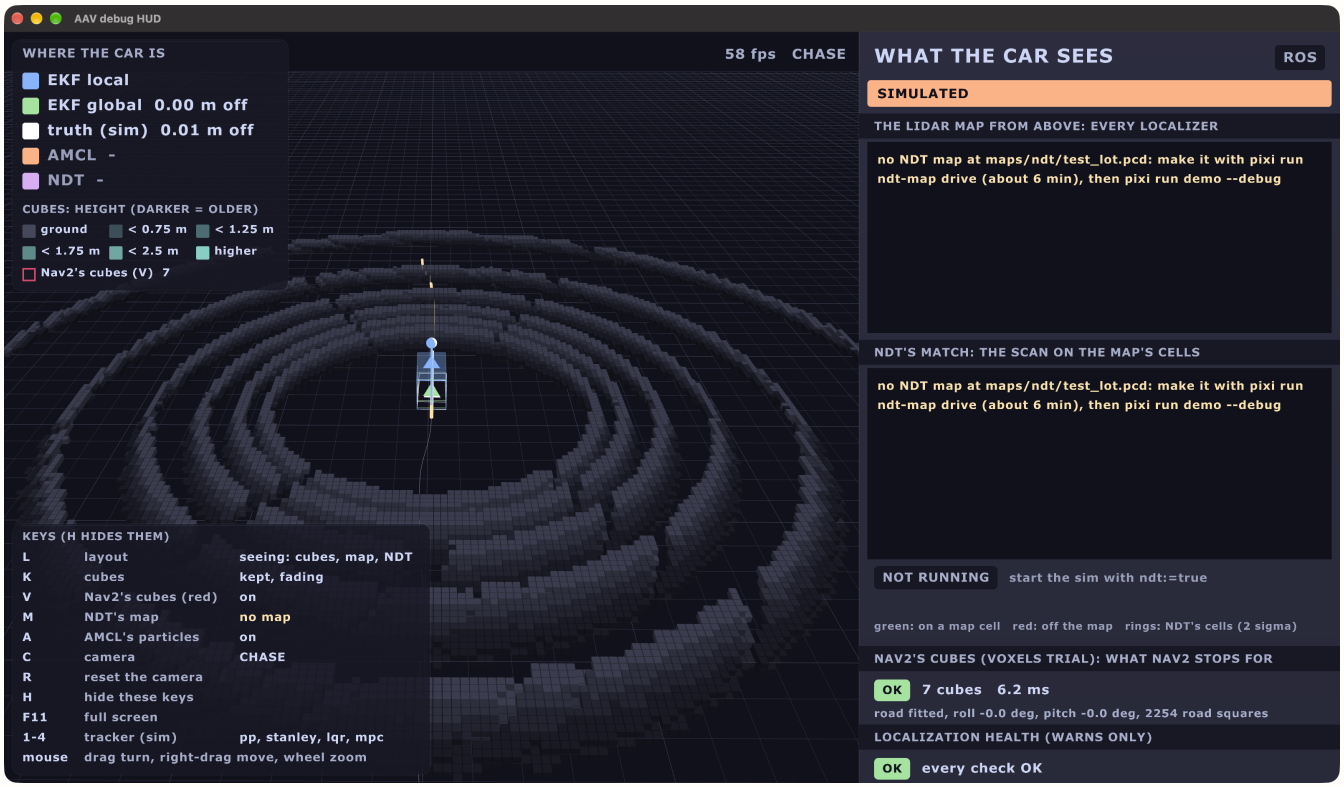
Open it, already connected: Foxglove.

Screen	0.2	0.3
Drive (the cockpit)	driving, Nav2, cubes, localizers, algorithms	absorbs the debug screen
Hardware (the panel)	sensors, buttons, steering's ends	stays separate

Screen	0.2	0.3
HUD and **map**	in the car	stay in the car

The debug screen

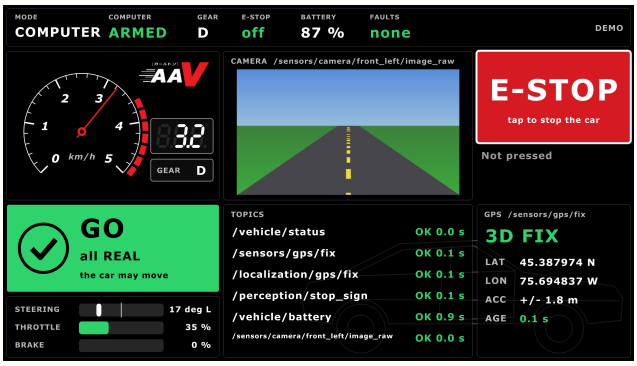
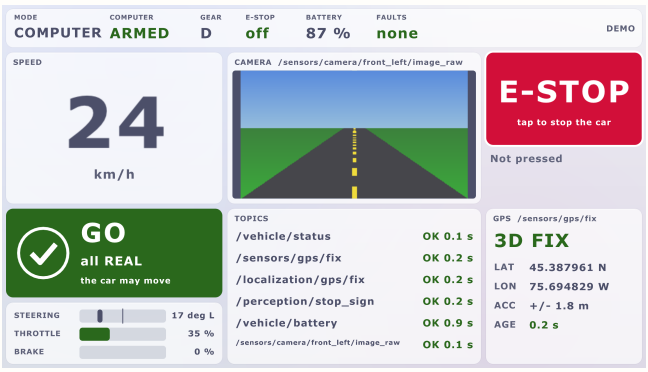
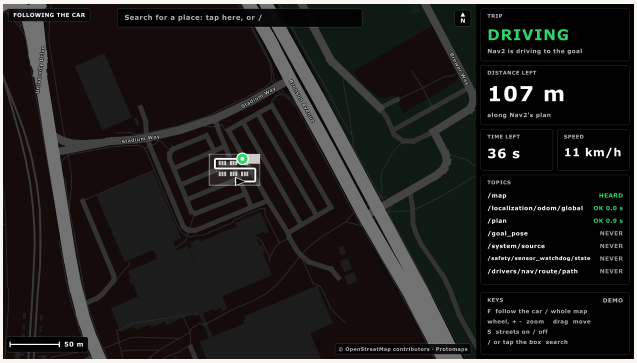
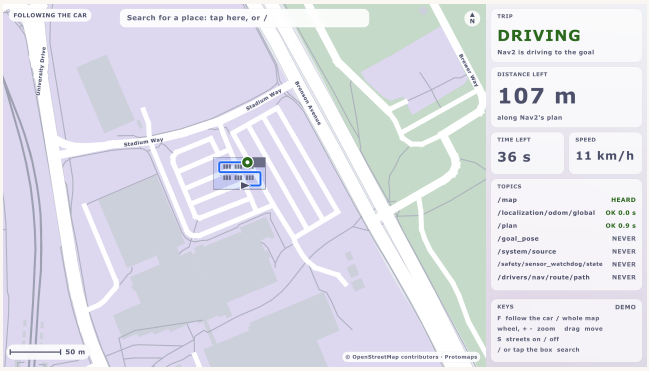
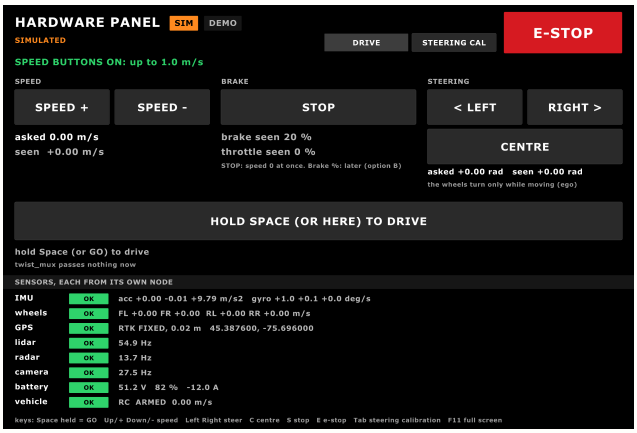
The lidar's cubes in 3D, where each localizer puts the car, Nav2's own cubes and the localization health checks. Live against Gazebo, in the new colours:



The screens: new colours

Ben, 1 Nov: "kind of ugly", too many primary colours. Then, 2 Nov: "something nicer than Catppuccin": **Initial D red and black, on every screen**. Every screen shares one theme, chosen in its yaml (`theme:`):

Theme	For
`initial_d` (the default)	the car, day or night: black, white, Carleton's red; sharp
`mocha`	dark and soft: night
`latte`	light: daytime
`calm`	Ben's colours from his Carleton work: **waits on Ben**

Initial D	Light (Latte)
	
	
	

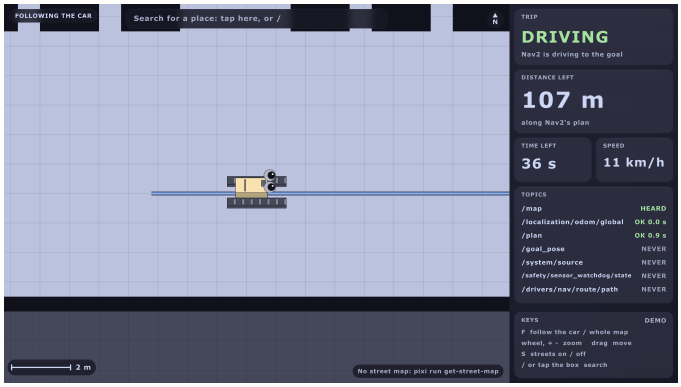
Mocha and Latte are Catppuccin's palettes, MIT licence. More: Screens.

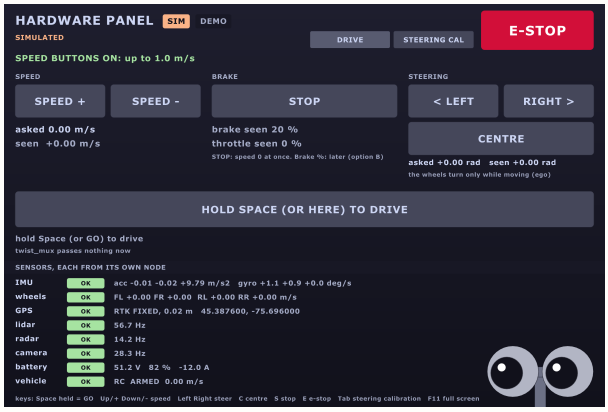
The little robot

A tribute to WALL-E, drawn by us: a boxy robot on treads, with two big round eyes.

Where	What it does
Map window (`car_marker: robot`)	the car is the robot, seen from above; its treads cover the car's true outline
Hardware panel, debug screen	after 60 s of quiet, the eyes peek over the bottom edge and blink
Hardware panel	it drives across at start-up, and after a lock-to-lock check passes

On the map	Peeking





It never covers a button or a line of words, and never shows while the car moves or anything warns. Turn it off with `easter_egg: false`. More: Screens.

Steering: find the ends and the centre

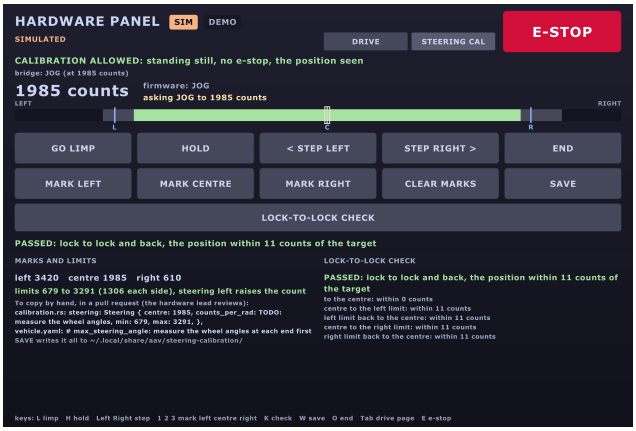
The steering pulls right: its centre is probably off. 0.2 can measure the steering actuator's two ends and its centre, then set **equal limits** on both sides.

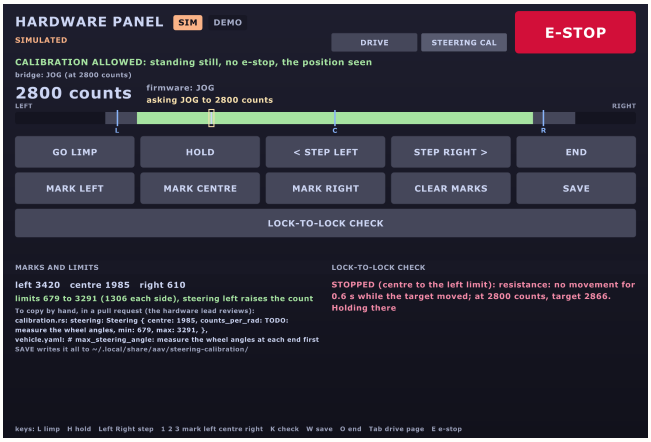
```
laptop (hardware panel)  ──►  Orin bridge  ──►  ESP32  ──►  steering actuator
     GO LIMP / STEP           link version 3     limp, or       (Jrk G2 board)
     MARK left/centre/right                      small steps
     ◄── the raw position (counts), 10 times a second ──┘
```

Step	What you do	What it does
1	**GO LIMP**	the motor lets go; you turn the wheels by hand

Step	What you do	What it does
2	or **STEP LEFT / RIGHT**	small steps, inside today's limits
3	**MARK** left, centre, right	it works out equal limits: the smaller side less 5 %
4	**LOCK-TO-LOCK CHECK**	slowly centre → left → centre → right → centre; **stops and holds** at any resistance
5	**SAVE**	writes the numbers to a file; a person copies them in a pull request

Passed	Stopped on a stiff spot





The safety rules, in the firmware and tested:

Rule	Why
Only while stopped, with the brake on	nothing else may move
Throttle held at zero	the same
Leaving limp holds where the wheels are	nothing snaps back
The Orin's heartbeat lost → calibration ends	a dropped laptop is safe
Never writes the firmware or `vehicle.yaml`	a person checks the numbers

On car day the first way is **Pololu's own program** over USB (Way A): today's C++ firmware can't do this. The panel's page (Way B) comes once the Rust firmware with link version 3 is on the car. More: Actuator tests.

Car day, in two days

Three reviewers read the plan and said: don't destroy the way back before the car has moved on our code. Ben agreed.

Day	What	Can you undo it?
Day A	back up every machine (checked); steering ends (Way A); brake on stands; the Rust ESP32 and the joystick on stands; walking pace once a named person signs; the car tests and the map	yes: nothing destroyed
Day B (later)	the Pis to Ubuntu 24.04; the Orin reflashed to JetPack 7.2; the lab desktop	only from the backups, so a restore is proven first

Every step you can't undo has a **sign-off line** (who, time). The people:

Role	Does
Hardware lead	decides on the hardware; OKs the ESP32 flash
E-stop person	holds the Kar-Tech stop, always
Safety driver	in the seat
Laptop person	runs the commands
Approver	signs the steps you can't undo

The runbook: Car day. Setting up one computer: Car setup.

`car_setup.sh`: one computer at a time

It does	Safe because
back up, upgrade, install, start at boot, check	it only prints the steps unless given `--apply` and the role typed in
`do-release-upgrade` (Ubuntu 22.04 → 24.04)	only with `--release-upgrade`; never on Pi OS or the Orin

It does	Safe because
STOPs	a pending reboot, the wrong computer, an upgrade too early

Know the car

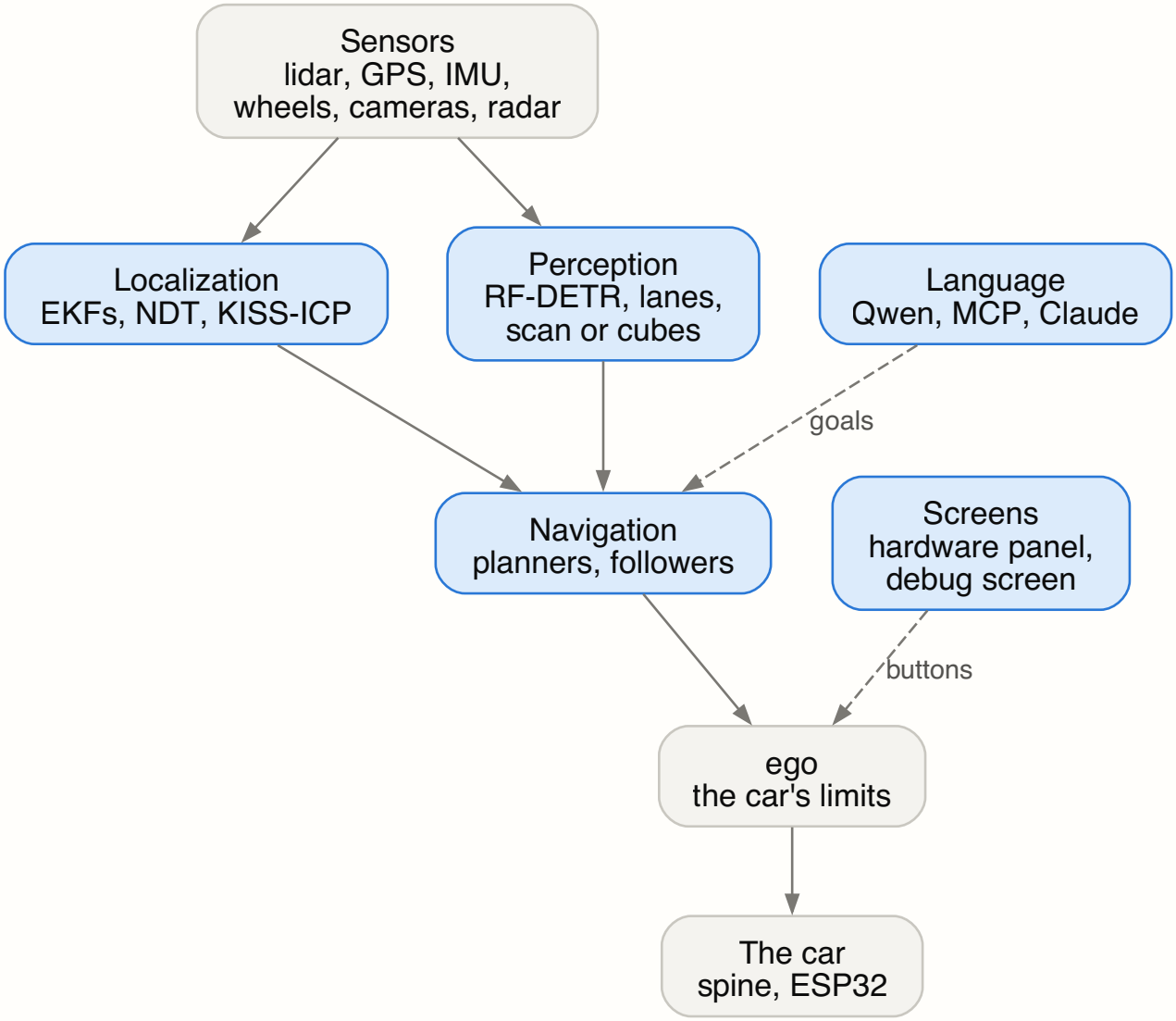
Tests that measure the car, so each drive makes it more exact. Each fixes one number in `vehicle.yaml`; repeat until the error is small.

Test	Fixes
Drive 10 m straight, tape it	the wheel size (today 0.36 m or 0.42 m: a 17 % speed error)
Speed step, 0 → 1 m/s	how fast it speeds up
Brake from 1 m/s	stopping distance
Coast down	drag
A circle at a fixed steering angle	steering gain and centre
Lock to lock	the steering limits

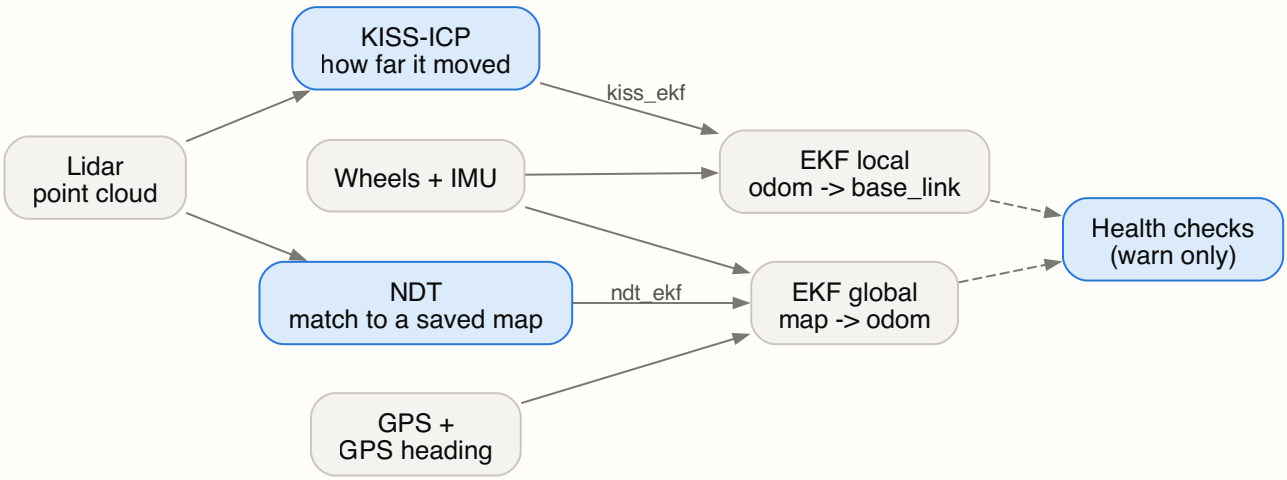
Record every session with the lidar on (`tools/record.sh --lidar`): the same drives build the lot's NDT map. More: Know the car.

How the car thinks: the algorithms

The same stack as in What's new. In plain words:

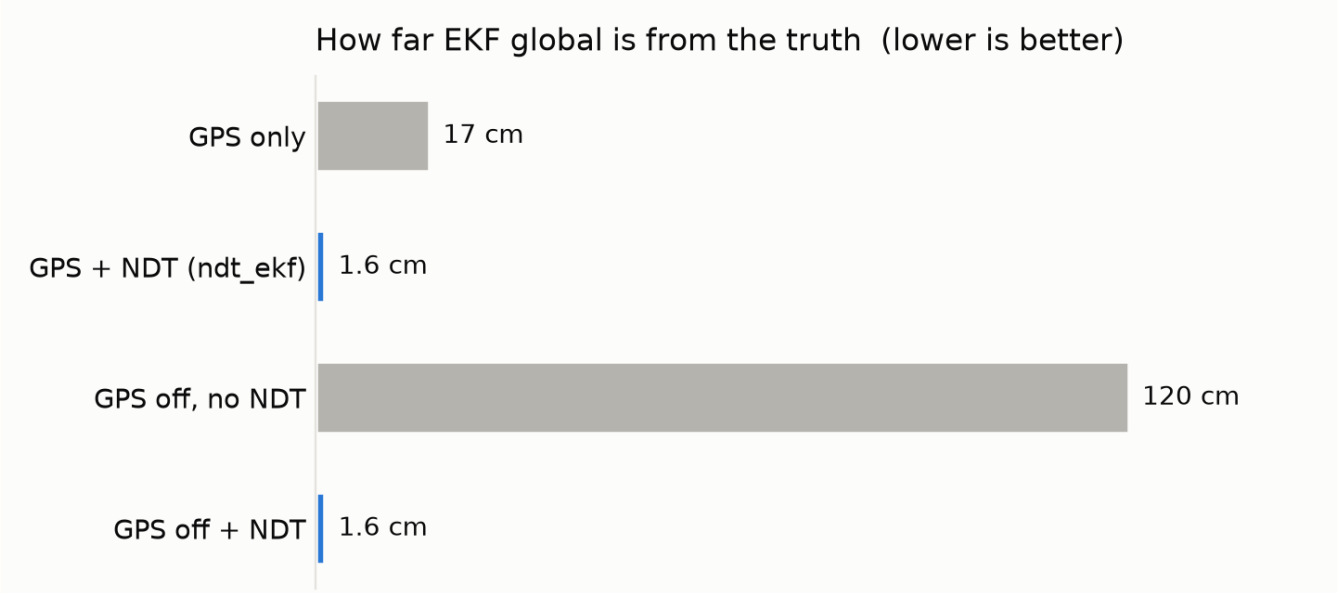


Where am I? (localization)

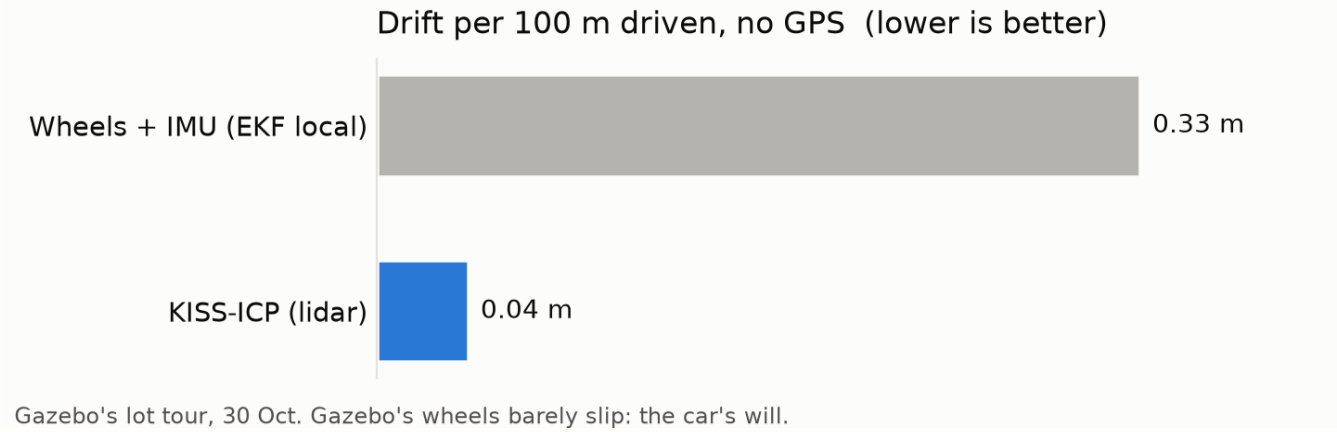


Part	In plain words	Default?
EKF local	blends the wheels and the IMU: smooth, never jumps, drifts slowly	yes
EKF global	adds the GPS (and NDT): knows where on the map, may jump a little	yes
NDT	matches the lidar's points to a saved map of the lot: "I'm here"	trial (`ndt_ekf:=true`)
KISS-ICP	matches each lidar scan to the last one: "I moved this far"	trial (`kiss_ekf:=true`)

Why two EKFs? They do talk, through TF. The GPS only goes into the global one, so the car's short-term motion never jumps when a GPS fix lands (ROS's REP-105).



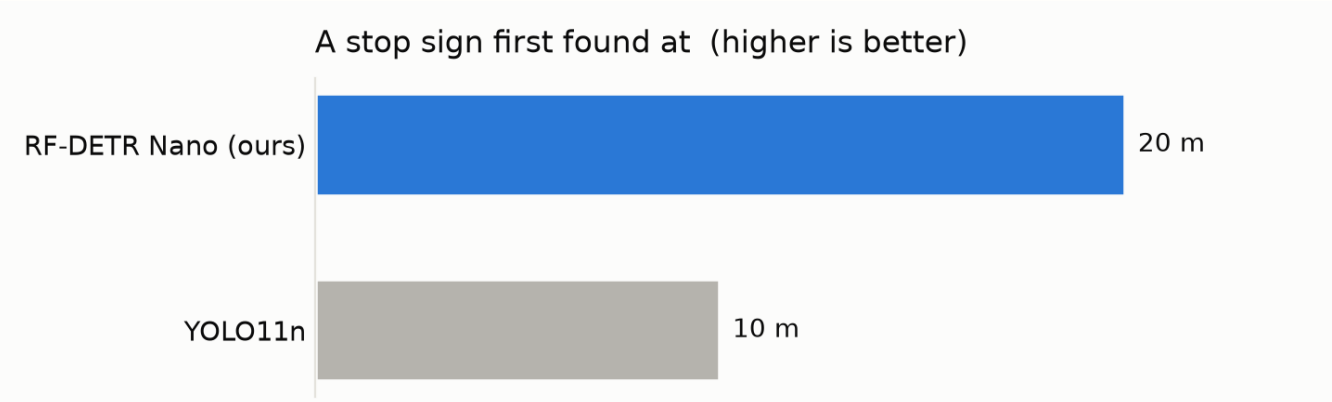
Gazebo's lot tour, 30 Oct.



Gazebo's lot tour, 30 Oct. Gazebo's wheels barely slip: the car's will.

What's around? (perception)

Part	In plain words
RF-DETR Nano	finds objects and stop signs in the camera (Apache-2.0, not YOLO)
OpenCV lanes	finds the lane lines
Lidar scan or cubes	turns the lidar's points into a flat scan, or into 3D cubes, for Nav2

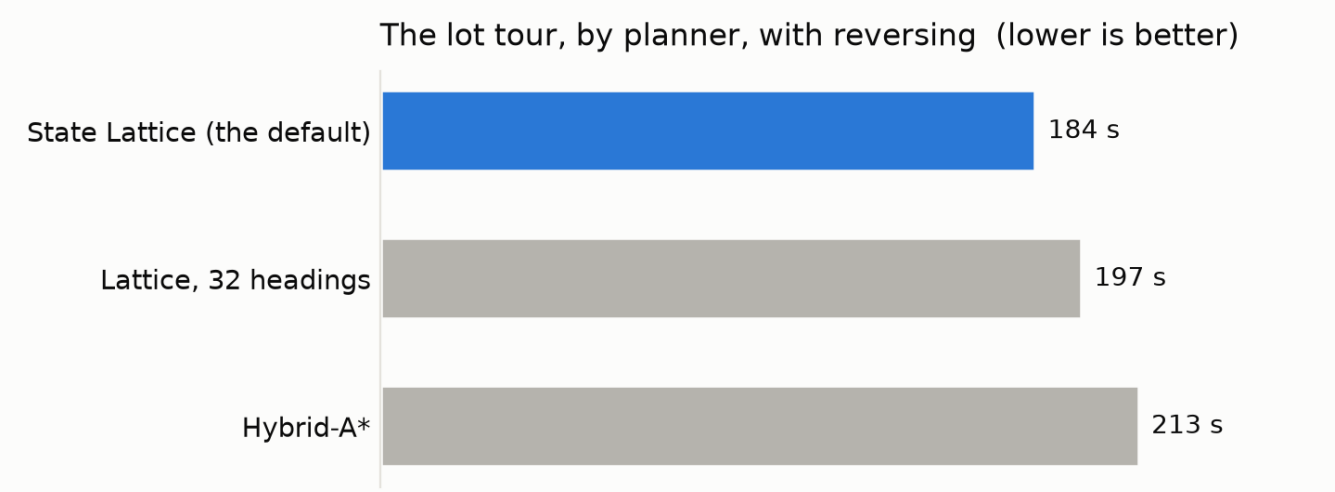


Gazebo road world, 20 Oct. RF-DETR: Apache-2.0, no false finds.

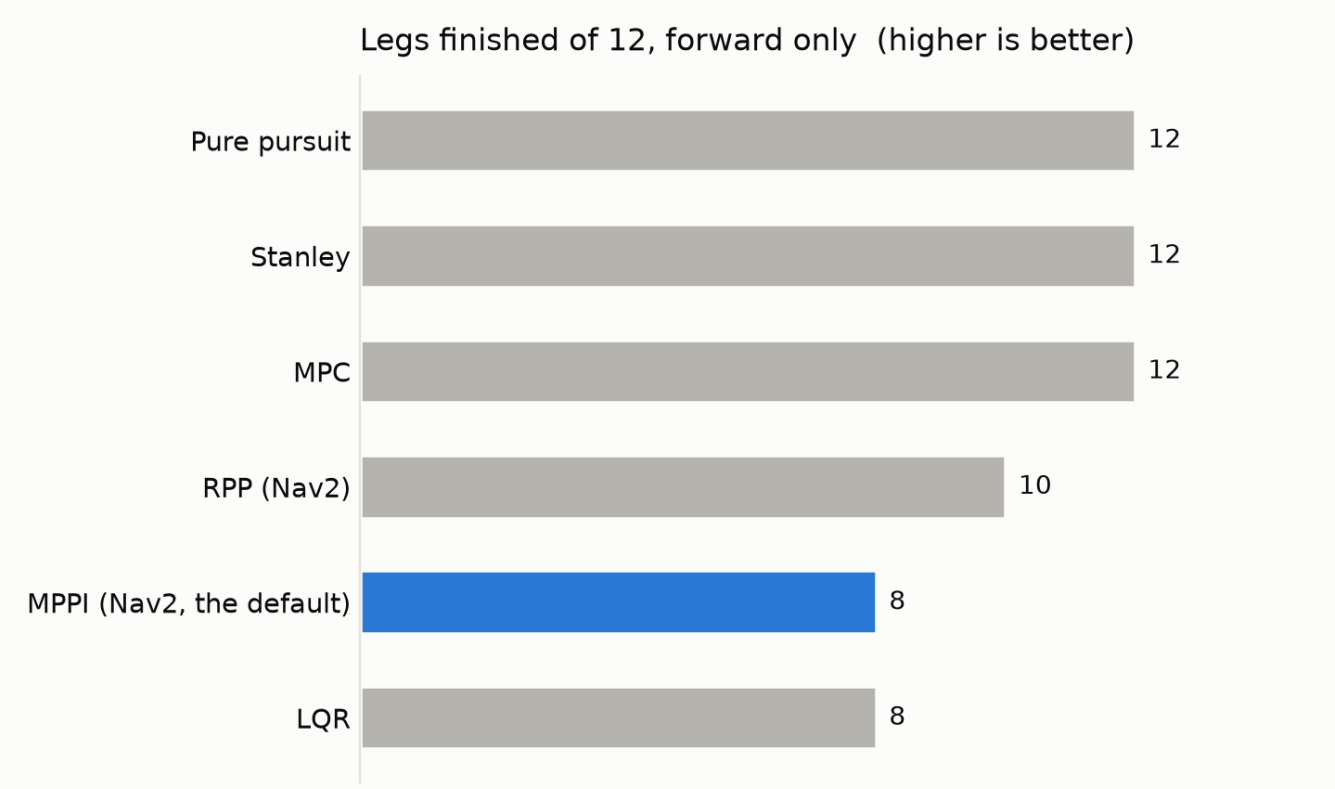
Getting there: planners and path followers

A **planner** draws the path; a **path follower** steers along it.

Part	In plain words	Default?
State Lattice	joins short moves the car can really make, forwards and backwards	yes, where it may reverse
Hybrid-A*	searches forward moves only	yes, forward only
MPPI	tries hundreds of short futures, keeps the best; reads the costmap	yes
RPP (Regulated Pure Pursuit)	aims at a point ahead on the path, slows in bends	trial



Gazebo, 30 Oct, MPPI; mean of 4 runs (lattice32: 3). Every planner finished every leg.



Gazebo's lot, 29 Oct, 3 runs of 4 legs each. The U-turn decided it. MPPI stays the default: it reads the costmap.

Trial: the diffusion planner

A small model that turns random noise into a path in a few steps, and offers 16 paths at once. Ben, 2 Nov: "Put it in 0.2". It's in, as a trial, off by default; the car refuses it.

Question	As built (2 Nov)
Where it plugs in	one more planner, `planner:=diffusion`, beside the lattice
What keeps it safe	every path checked (G1-G8: collisions, the car's turns, headings, time); if none pass or it's late, the lattice's path
What it learned from	45,092 lattice and Hybrid-A* paths on 791 maps; 3.9 M weights, 20 min on the Mac
What runs it	a Python node; the model in ONNX Runtime (CPU, 75 ms)
Gazebo, lot tour, 2 runs	8 of 8 legs, 178-182 s (lattice 186-188 s); 63 % of plans from the model, 37 % the lattice's

Not a winner yet: the design asks for 90 % from the model. The U-turn is its weakest leg. Results: trials; the design: Diffusion planner.

Kept for 0.3 and later

Every idea is kept (Ben, 1 Nov: "I do like the 23 ideas? lets keep them around").

Idea	State
Car day itself (0.3 = it drove)	runbook ready
The diffusion planner: 90 % of plans from the model, the U-turn, the Orin	trial in 0.2
The talking model: explains, never drives	idea
The Initial D HUD: the arcade gauge in Carleton red and black, our own logo	in 0.2: on every screen (2 Nov)
The little robot on the in-car HUD	after the Initial D HUD
Calm colours	waits on Ben
The cockpit's algorithm switch; one screen from the cockpit and the debug screen	in 0.2: the cockpit shows which algorithms run (3 Nov); the switch and the debug screen's views for 0.3
More hardware panel pages: measurements, joystick, the car tests	idea

Idea	State
Python speed on the Orin (the cubes to Rust if too slow)	measure on car day
Brake and throttle: measure, then set	after the steering
Day B: Ubuntu 24.04 on the Pis, JetPack 7.2 on the Orin	after a restore is proven

The list: TODO, "0.3 and later".

How solid is 0.2?

Solid	every test green; every profile runs in Gazebo; the screens seen live against it
Untested	anything on the real car: the steering page, `car_setup.sh --apply`, the boot service, the joystick path
Open	the joystick-alive check on the real pad (A9b), the throttle cap, the people, the hardware lead's answers

The checks run for this page

Run on Ben's Mac, 2 November, for this page:

Check	Result	What's left
`pixi run test`	✅ 1516 tests, 0 failures	nothing
`bash tools/car_setup.sh <role>`, all six roles (dry run)	✅ no STOP except the two planned refusals (`motor_pi` and `localization` won't upgrade yet)	nothing
`pixi run car-day-check`	⬜ 7 OK, **11 missing**, 7 to tick by hand	mostly downloads not yet on this Mac: `get models`, `get mediamtx`, `esp tools`, `esp build`, `esp qemu-tools`, `esp qemu-check`, `car-day-build` (the C++ way back), `docs` (the PDF)

Check	Result	What's left
`pixi run esp qemu-check`	⬜ skipped: the ESP32's Rust and QEMU aren't installed here	`pixi run esp tools`, then `esp qemu-tools`, then run it again

The ESP32 tools install Espressif's Rust on the laptop, so they wait for Ben's OK.